

Stationary Multihomogeneous Poisson Particle Systems

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Abstract

With motivation from Molchanov and Stucki [1] we study Poisson particle systems generated by componentwise scaling of vector-valued random processes. We characterise stationarity in terms of the generating process. For finite mixtures of log-Gaussian and log-Lévy marks on the real line, we show that the mixture system is stationary if and only if (iff) every component system is stationary. We also construct stationary systems from clusters and establish a converse representation under suitable integrability assumptions. These representations yield formulas for the maximal index, with applications to multivariate Pickands constants.

Keywords: Poisson particle systems; multihomogeneous measures; stationarity; finite mixtures; cluster representations; Pickands constants; maximal index.

1 Introduction

Fix $q \in \mathbb{N}$ and put

$$G = (0, \infty)^q, \quad \chi(r) = \prod_{i=1}^q r_i, \quad \vartheta(dr) = \prod_{i=1}^q r_i^{-2} dr_i.$$

We equip G with componentwise multiplication and let

$$\tilde{\Pi} = \sum_{j \geq 1} \delta_{R_j} \sim \text{PRM}(\vartheta)$$

be a Poisson random measure on G , where $R_j = (R_{j1}, \dots, R_{jq}) \in G$ are its atoms. Independently of $\tilde{\Pi}$, let $Z^{(j)}$, $j \geq 1$, be independent copies of an $\mathbb{R}^q$ -valued random process $Z = (Z(t))_{t \in \mathbb{T}}$, where $\mathbb{T} = \mathbb{R}$. We call Z the generic mark and $Z^{(j)}$ the mark attached to R_j . All these random objects are defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The j th particle follows the trajectory

$$F_j(t) = R_j \odot Z^{(j)}(t), \quad t \in \mathbb{T},$$

where

$$r \odot x = (r_1 x_1, \dots, r_q x_q), \quad r \in G, \quad x \in \mathbb{R}^q.$$

At time $t \in \mathbb{T}$ the corresponding particle system of Z is defined as

$$\mathcal{N}_Z(t) = \sum_{j \geq 1} \delta_{R_j \odot Z^{(j)}(t)}. \tag{1}$$

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To retain the identity of each particle at several time points $t_1, \dots, t_n \in \mathbb{T}$ consider further

$$\mathcal{N}_Z(t_1, \dots, t_n) = \sum_{j \geq 1} \delta_{(F_j(t_1), \dots, F_j(t_n))}.$$

These sums are understood as extended counting measures with laws defined on the σ -field generated by the counting maps. Of particular interest are finite-dimensionally stationary $\mathcal{N}_Z$'s, i.e., for all $n \geq 1$ and $u, t_1, \dots, t_n \in \mathbb{T}$

$$\mathcal{N}_Z(t_1 + u, \dots, t_n + u) \stackrel{d}{=} \mathcal{N}_Z(t_1, \dots, t_n),$$

where $\stackrel{d}{=}$ denotes equality in distribution. This construction is the componentwise multiplicative analogue of the additive particle systems studied in [2, 1]. For instance, in [1, Sec. 2] particles have trajectories

$$x_j + \xi^{(j)}(t), \quad x_j \in \mathbb{R}^q, \quad t \in \mathbb{R}.$$

On strictly positive trajectories, componentwise logarithms transform our construction into this additive form, with the points $\log R_j$ having intensity

$$e^{-\langle \mathbf{1}, x \rangle} dx, \quad x \in \mathbb{R}^q,$$

where $\langle \cdot, \cdot \rangle$ is the standard inner product in $\mathbb{R}^q$ and $\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^q$. The system $\mathcal{N}_Z$ may be stationary even when Z is not.

Put $\mathcal{X} = (\mathbb{R}^q)^\mathbb{T}$ and equip it with $\mathcal{E}_{\text{cyl}} = \sigma\{f \mapsto f(t) : t \in \mathbb{T}\}$. For $r \in G$ and $u \in \mathbb{T}$ set

$$(S_r f)(t) = r \odot f(t), \quad (B^u f)(t) = f(t - u).$$

A nonzero measure ν on $(\mathcal{X}, \mathcal{E}_{\text{cyl}})$ is called χ -homogeneous if

$$\nu(S_r A) = \chi(r)^{-1} \nu(A), \quad r \in G, \quad A \in \mathcal{E}_{\text{cyl}}, \quad (2)$$

which is abbreviated as $\nu \in M(\mathcal{X})$; we write $\nu \in M_\sigma(\mathcal{X})$ when ν is also σ -finite. The same notation will be used for measures on the regular cylindrical path carriers $(\mathcal{C}, \mathcal{E}_{\mathcal{C}})$ introduced in Definition 2.2 below.

Our primary object is the *multihomogeneous Poisson particle system* (mPps)

$$\Pi_\nu = \sum_{j \geq 1} \delta_{F_j} \sim \text{PRM}(\nu), \quad \nu \in M_\sigma(\mathcal{X}). \quad (3)$$

All Poisson laws are understood on the counting-map σ -field. Whenever an atomic sum is displayed, we choose a proper Poisson realisation without changing its law; see [3, Thm. 3.6 and Cor. 3.7]. Nonzero homogeneity forces $\nu(\mathcal{X}) = \infty$, so Π_ν has infinitely many particles almost surely. Multiplicities are retained and no argument depends on the enumeration.

For an $(\mathcal{X}, \mathcal{E}_{\text{cyl}})$ -valued random element Z , define the corresponding intensity measure

$$\nu_Z(A) = \mathbb{E} \left\{ \int_G \mathbb{I}_{\{S_r Z \in A\}} \vartheta(dr) \right\}, \quad A \in \mathcal{E}_{\text{cyl}}. \quad (4)$$

We call Z a *stochastic representer* of a measure ν on $(\mathcal{X}, \mathcal{E}_{\text{cyl}})$ when $\nu = \nu_Z$. When ν_Z is σ -finite, the marking and mapping theorems [3, Thms. 5.1 and 5.6] imply

$$\sum_{j \geq 1} \delta_{S_{R_j} Z^{(j)}} \stackrel{d}{=} \Pi_{\nu_Z}.$$

Thus the particle-system law depends on Z only through ν_Z .

Here and below $(T_{\#}\mu)(A) = \mu(T^{-1}A)$ denotes the pushforward of a measure μ under a measurable map T .

The simplest case is $Z = (Z_1, \dots, Z_q)$ with independent coordinates. Let $\mathcal{X}_1 = \mathbb{R}^{\mathbb{T}}$ have its cylindrical σ -field and identify $\mathcal{X}$ with $\mathcal{X}_1^q$. If every measure ν_{Z_i} is σ -finite, then

$$\nu_Z = \bigotimes_{i=1}^q \nu_{Z_i}.$$

If every ν_{Z_i} is shift-invariant, then

$$(B^u)_{\#}\nu_Z = \bigotimes_{i=1}^q (B^u)_{\#}\nu_{Z_i} = \bigotimes_{i=1}^q \nu_{Z_i} = \nu_Z, \quad u \in \mathbb{T}.$$

It turns out that (see Theorem 3.1 below)

$$(B^u)_{\#}\Pi_{\nu} \stackrel{d}{=} \Pi_{\nu} \iff (B^u)_{\#}\nu = \nu, \quad u \in \mathbb{T}. \quad (5)$$

We call the mPps Π_{ν} stationary when the identity on the left holds for every $u \in \mathbb{T}$. In particular it implies the finite-dimensional stationarity of

$$(\pi_{t_1, \dots, t_n})_{\#}\Pi_{\nu}, \quad \pi_{t_1, \dots, t_n}(f) = (f(t_1), \dots, f(t_n)).$$

By [4, Thm. 2.2], every $\nu \in M_{\sigma}(\mathcal{X})$ has a stochastic representer. For $\nu \in M_{\sigma}(\mathcal{C})$ on a regular carrier, a representer can be chosen in $\mathcal{C}$ and is jointly measurable as a field; see Lemma 2.3. The stationarity identities are cylindrical. Regular carriers provide measurable compact windows and path integrals, while the log-Gaussian and maximal-index results use continuous paths. Henceforth $\mathbb{T}$ is a locally compact second-countable Hausdorff (lcsH) group with identity e , shifts (9) and $\nu = \nu_Z$.

Q1: *How can stationarity of Π_{ν} be recognised from a stochastic representer Z of $\nu \in M_{\sigma}(\mathcal{X})$?*

Theorem 3.1 characterises stationarity by shift invariance of ν and the identity (25) for homogeneous functionals. The main examples considered below for which the stationary criteria can be checked directly are:

- (i) Matrix transforms $Z = e^{A \log Z^*}$, $A \in \mathbb{R}^{q \times q}$, componentwise, of strictly positive marks with $Z^*(e) = \mathbf{1}$. If ν_{Z^*} is shift-invariant and $A^{\top} \mathbf{1} = \mathbf{1}$, then ν_Z is shift-invariant, see Example 3.4; Proposition 3.3 gives an exact criterion for a fixed input.
- (ii) Log-Lévy marks Z specified in Equation (32) below; Proposition 3.5 characterises shift invariance; the vector coordinates need not be independent.
- (iii) Log-Gaussian marks, i.e., strictly positive marks whose logarithms are Gaussian fields already investigated in [2, 1]. For instance, for $\mathbb{T} = (\mathbb{R}, +)$ and $\kappa \in (0, 2]$ set

$$Z_i(t) = \exp \left\{ Y_{\kappa, V, i}(t) - (S_{\kappa, V}(t) \mathbf{1})_i \right\}, \quad i = 1, \dots, q, \quad (6)$$

where $V \in \mathbb{R}^{q \times q}$ is such that

$$R_{\kappa, V}(t, s) = S_{\kappa, V}(t) + S_{\kappa, V}(-s) - S_{\kappa, V}(t - s), \quad s, t \in \mathbb{T}$$

is the covariance kernel of a continuous centred Gaussian process $Y_{\kappa, V}(t), t \in \mathbb{T}$, where

$$S_{\kappa, V}(t) = |t|^{\kappa} (V \mathbb{I}_{\{t \geq 0\}} + V^{\top} \mathbb{I}_{\{t < 0\}}), \quad t \in \mathbb{T}.$$

Then in view of [1] ν_Z is shift-invariant since (37)–(38) below are valid.

- (iv) Product-cluster marks. Example 5.6 starts from a deterministic bivariate product cluster and yields a continuous non-log-Gaussian representer with shift-invariant intensity and explicit simultaneous-hit intensities.

For $q = 1$ and $\mathsf{T} = \mathbb{R}$, a jointly measurable Z with shift-invariant ν_Z and $\mathbb{E}\{|Z(0)|\} < \infty$ is L^1 -continuous, see [5]. For nonnegative spectral processes in the max-stable setting see also [6, 7]. The following question for $q > 1$ and T a general lscH group arises naturally:

Q2: *What regularity properties of Z are forced by stationarity of Π_ν ?*

In the light of Theorem 3.7 below, stationarity of Π_ν implies stochastic continuity of every jointly measurable representer Z of a σ -finite shift-invariant cylindrical intensity ν_Z .

A finite mixture of marks with shift-invariant intensities has a shift-invariant intensity.

For log-Gaussian and log-Lévy marks we ask:

Q3: *When does stationarity of Π_{ν_Z} , where Z is a finite mixture of representers Z_k , imply stationarity of every component system $\Pi_{\nu_{Z_k}}$?*

For continuous log-Gaussian marks Theorem 4.1 presents an exact criterion in terms of permutations of the canonically normalised weighted component laws, after merging coincident laws. On a connected index group, stationarity holds iff every component system is stationary. On $\mathsf{T} = \mathbb{R}$, the same componentwise conclusion holds for finite mixtures of log-Gaussian and log-Lévy marks, including matrix transforms of log-Lévy inputs.

For $q = 1$, cluster random fields yield shift-invariant tail measures and mixed moving maxima representations; see e.g., [8, 9, 10]. The role of cluster random fields in our general setting leads to the following question:

Q4: *How can stationary multihomogeneous particle systems be constructed and represented by clusters?*

Section 5 constructs stationary systems from product clusters. Proposition 5.4 reconstructs a cluster from a $\mathcal{C}$ -valued representer satisfying (46) whose intensity is shift-invariant and satisfies (19).

Scalar cluster fields give representations of classical Pickands constants, see [11, 10]. Vector-valued Gaussian extremes lead to related constants appearing in [12]. For Z in (6) with $c := \mathbf{1}^\top V \mathbf{1} > 0$ a limiting constant of the form studied in the aforementioned contribution, up to normalisation, is given by (vector inequalities are understood componentwise)

$$\mathcal{H} := \lim_{S \rightarrow \infty} \frac{1}{S} \int_{\mathbb{R}^q} e^{\langle \mathbf{1}, x \rangle} \mathbb{P} \{ \exists t \in [0, S] : Y_{\kappa, V}(t) - S_{\kappa, V}(t) \mathbf{1} > x \} dx. \quad (7)$$

In view of the maximal-index framework of [8, Section 2.6] and [11] for $q = 1$ Pickands constants are closely related to maximal indices. We formulate now the following question for $q \geq 1$ and T an amenable unimodular group:

Q5: *Can the maximal index be expressed directly through a stochastic representer or a product cluster?*

The answer is affirmative and our result shows that the maximal index is the limiting hit intensity per Haar volume along a suitable Folner sequence. Corollary 6.2 specialises them to the Gaussian constant in (7).

Organisation of the paper: Section 2 introduces cylindrical measures, a practical sigma-finiteness criterion, path carriers and stochastic representers. Section 3 treats stationarity and automatic continuity, Section 4 finite log-Gaussian and log-Lévy mixtures and Section 5 product clusters and reconstruction, with a non-Gaussian example. Section 6 presents the maximal-index representations and their Gaussian specialisation. Auxiliary lemmas and proofs are collected in Section 7.

2 Preliminaries

In this section we discuss cylindrical measures, σ -finiteness, regular carriers and stochastic representers. We also introduce compact windows which define important homogeneous maps. For notational simplicity, hereafter we do not consider characters $\chi_\alpha(r) = \prod_{i=1}^q r_i^{\alpha_i}$ with $\alpha = (\alpha_1, \dots, \alpha_q) \in (0, \infty)^q$; since these can be reduced to χ by componentwise signed-power changes of the trajectories and the corresponding power changes of the radial variables.

2.1 Cylindrical measures and σ -finiteness

Let $\mathbb{T}$ be an lscH group with identity e , set

$$\mathcal{X} = (\mathbb{R}^q)^\mathbb{T}, \quad \mathcal{E}_{\text{cyl}} = \sigma\{\pi_t : t \in \mathbb{T}\}, \quad \pi_t(f) = f(t), \quad f \in \mathcal{X} \quad (8)$$

and define

$$(S_r f)(t) = r \odot f(t), \quad (B^a f)(t) = f(a^{-1}t). \quad (9)$$

The scaling map $(r, f) \mapsto S_r f$ is jointly measurable into $(\mathcal{X}, \mathcal{E}_{\text{cyl}})$, since each of its evaluations is $r \odot f(t)$. For fixed $a \in \mathbb{T}$, the shift B^a is a measurable bijection with measurable inverse $B^{a^{-1}}$. Thus (4) defines a measure for every $(\mathcal{X}, \mathcal{E}_{\text{cyl}})$ -valued random element Z , without assuming joint measurability of $(t, \omega) \mapsto Z(t, \omega)$. For $f \in \mathcal{X}$ set

$$\mathfrak{W}_f(t) = \prod_{i=1}^q |f_i(t)|, \quad t \in \mathbb{T}. \quad (10)$$

We call $\mathfrak{W}_f$ the common-character field of f .

When $\mathbb{T}$ is uncountable, $\mathcal{E}_{\text{cyl}}$ is not countably generated and universal evaluation on $\mathbb{T} \times \mathcal{X}$ is not jointly measurable. In particular, path integrals, uncountable suprema and pathwise nonzero-component events need not be cylinder measurable. Nevertheless, sigma-finiteness and homogeneity provide the following countable detection property.

Lemma 2.1. *Let $E = \mathcal{X}$ or a regular cylindrical path carrier as in Definition 2.2, equipped with its cylindrical or trace cylindrical σ -field, respectively. If $\nu \in M_\sigma(E)$, there is a nonempty countable set $D \subset \mathbb{T}$ such that*

$$V_D = \bigcap_{i=1}^q \bigcup_{h \in D} \{f \in E : f_i(h) \neq 0\}$$

is measurable and ν -conull. If $\nu = \nu_Z$, then $\mathbb{P}\{Z \in V_D\} = 1$.

Since σ -finiteness is used throughout, we give next a simple criterion for $\nu_Z \in M_\sigma(\mathcal{X})$ when Z is an $(\mathcal{X}, \mathcal{E}_{\text{cyl}})$ -valued random element. First note that since $\vartheta(G) = \infty$, every measurable set A in $\mathcal{X}$, or in a regular carrier such that $S_r A = A$ for all $r \in G$ satisfies

$$\nu_Z(A) = 0 \iff \mathbb{P}\{Z \in A\} = 0. \quad (11)$$

Now suppose that for a countable $D \subset \mathbb{T}$

$$\mathbb{P}\{\text{for } i = 1, \dots, q \text{ there is } h \in D \text{ with } Z_i(h) \neq 0\} = 1 \quad (12)$$

and for all $\mathbf{h} = (h_1, \dots, h_q) \in D^q$

$$p_{\mathbf{h}} := \mathbb{E}\left\{\prod_{i=1}^q |Z_i(h_i)|\right\} < \infty. \quad (13)$$

Since V_D^c is scale invariant, (12) and (11) yield $\nu_Z(V_D^c) = 0$. The countable family

$$A_{\mathbf{h}}^{(n)} = \{f \in \mathcal{X} : |f_i(h_i)| > 1/n, \ i = 1, \dots, q\}, \quad \mathbf{h} \in D^q, \quad n \geq 1$$

covers V_D , and radial integration implies $\nu_Z(A_{\mathbf{h}}^{(n)}) = n^q p_{\mathbf{h}} < \infty$ and hence $\nu_Z \in M_{\sigma}(\mathcal{X})$. In particular, a single anchor $D = \{e\}$ suffices whenever $\mathbb{P}\{\mathfrak{W}_Z(e) > 0\} = 1$ and $\mathbb{E}\{\mathfrak{W}_Z(e)\} < \infty$; this includes $Z_i(e) = 1$ almost surely for $i = 1, \dots, q$. Condition (12) is necessary by Lemma 2.1, but (13) is not.

2.2 Regular carriers and compact windows

For path integrals and compact windows we use the following carriers.

Definition 2.2. A regular cylindrical path carrier over $\mathbb{T}$ is a set $\mathcal{C} \subset \mathcal{X}$ which is stable under all shifts B^a and componentwise scalings S_r , is equipped with the trace cylindrical σ -field

$$\mathcal{E}_{\mathcal{C}} := \mathcal{E}_{\text{cyl}}|_{\mathcal{C}} = \{A \cap \mathcal{C} : A \in \mathcal{E}_{\text{cyl}}\} = \sigma\{\pi_t|_{\mathcal{C}} : t \in \mathbb{T}\} \quad (14)$$

and has the following properties.

(a) The evaluation map

$$\text{ev} : (\mathbb{T} \times \mathcal{C}, \mathcal{B}(\mathbb{T}) \otimes \mathcal{E}_{\mathcal{C}}) \longrightarrow (\mathbb{R}^q, \mathcal{B}(\mathbb{R}^q)), \quad \text{ev}(t, f) = f(t)$$

is measurable.

(b) For every nonempty compact $K \subset \mathbb{T}$, there is a nonempty countable set $D_K \subset K$ chosen independently of f such that

$$\rho_{K,i}(f) := \sup_{t \in D_K} |f_i(t)| = \sup_{t \in K} |f_i(t)| < \infty, \quad f \in \mathcal{C}, \quad i = 1, \dots, q.$$

Fix such sets D_K . Their countability makes every $\rho_{K,i}$ $\mathcal{E}_{\mathcal{C}}$ -measurable. Condition (a) and stability make the actions in (9) jointly measurable on $\mathcal{C}$, with measurable inverses S_{r-1} and B^{a-1} for fixed parameters.

By condition (a) and the countable-support property of generated σ -fields, the inverse images under evaluation of a countable base of $\mathbb{R}^q$ all belong to $\mathcal{B}(\mathbb{T}) \otimes \sigma\{\pi_t|_{\mathcal{C}} : t \in D_0\}$ for one countable $D_0 \subset \mathbb{T}$. Taking sections at each $t \in \mathbb{T}$ gives

$$\mathcal{E}_{\mathcal{C}} = \sigma\{\pi_t|_{\mathcal{C}} : t \in D_0\}. \quad (15)$$

The map $f \mapsto (f(t))_{t \in D_0}$ is injective because (15) generates $\mathcal{E}_{\mathcal{C}}$, which separates points. Scaling a nonzero coordinate by two changes the path, so injectivity also implies

$$f_i \not\equiv 0 \iff f_i(t) \neq 0 \text{ for some } t \in D_0, \quad i = 1, \dots, q. \quad (16)$$

Fix one such D_0 throughout the rest of the paper.

The continuous path space $C(\mathbb{T}, \mathbb{R}^q)$ is a basic example: any countable dense subset of K is norming. The definition also allows discontinuous path classes as considered in [13].

For $\mathbb{T} = (\mathbb{R}, +)$, the càdlàg path space $D(\mathbb{R}, \mathbb{R}^q)$ is another example, with $D_0 = \mathbb{Q}$. For a compact K , add to a countable dense subset of K all its right-isolated points

$$R_K = \{t \in K : K \cap (t, t + \varepsilon) = \emptyset \text{ for some } \varepsilon > 0\}.$$

These points are countable, since their right gaps are disjoint. The resulting D_K is norming by right-continuity, and càdlàg paths are locally bounded. Joint evaluation in the two examples follows by approximating t with Borel maps of countable range: use the first point of a dense sequence within distance $1/n$ in a compatible metric for continuous paths, and $s_n(t) = 2^{-n}(\lfloor 2^n t \rfloor + 1) \downarrow t$ for càdlàg paths, then $\lim_{n \rightarrow \infty} f(s_n(t)) = f(t)$. The skeleton D_0 and the norming sets D_K serve different purposes; in particular, $D_0 \cap K$ need not be norming for discontinuous paths.

For every non-negative Borel $H : \mathbb{T} \times \mathbb{R}^q \rightarrow [0, \infty]$ and left Haar measure λ , condition (a) and Tonelli's theorem make the map

$$f \mapsto \int_{\mathbb{T}} H(t, f(t)) \lambda(dt)$$

$\mathcal{E}_{\mathcal{C}}$ -measurable.

On $(\mathcal{C}, \mathcal{E}_{\mathcal{C}})$, use $M(\mathcal{C})$, $M_{\sigma}(\mathcal{C})$ and ν_Z with the same definitions. For the inclusion $\iota : \mathcal{C} \rightarrow \mathcal{X}$, $\iota_{\#} \nu_Z$ is the cylindrical intensity of the same field. Since every $\mathcal{E}_{\mathcal{C}}$ -set is a cylindrical trace, equality and fixed-shift invariance of carrier measures are equivalent to those of their pushforwards.

Next, for a $\mathcal{C}$ -valued random element Z , consider the condition

$$\mathbb{P}\{Z_i \neq 0 \text{ for } i = 1, \dots, q\} = 1. \quad (17)$$

The event in (17) is measurable by (16).

Choose a cofinal increasing exhaustion $(K_n)_{n \geq 1}$ of $\mathbb{T}$ by nonempty compact sets and write

$$\rho_{n,i} = \rho_{K_n,i},$$

with $\rho_{K,i}$ as in Definition 2.2(b). For $\mathbf{n} = (n_1, \dots, n_q) \in \mathbb{N}^q$ and $\varepsilon \in (0, \infty)^q$ define the product window

$$U_{\mathbf{n}, \varepsilon} = \bigcap_{i=1}^q \{f : \rho_{n_i,i}(f) > \varepsilon_i\}. \quad (18)$$

For a measure ν on $(\mathcal{C}, \mathcal{E}_{\mathcal{C}})$ suppose that

$$\nu(U_{\mathbf{n}, \varepsilon}) < \infty, \quad \mathbf{n} \in \mathbb{N}^q, \quad \varepsilon \in (0, \infty)^q. \quad (19)$$

The identity $\int_0^\infty \mathbb{I}_{\{r\rho > 1\}} r^{-2} dr = \rho$, $\rho \geq 0$ and coordinatewise radial integration yield

$$\nu_Z(U_{\mathbf{n}, \varepsilon}) = \mathbb{E} \left\{ \prod_{i=1}^q \frac{\rho_{n_i,i}(Z)}{\varepsilon_i} \right\}, \quad \mathbf{n} \in \mathbb{N}^q, \quad \varepsilon \in (0, \infty)^q. \quad (20)$$

Since (K_n) is increasing (19) for $\nu = \nu_Z$ is equivalent to

$$\mathbb{E} \left\{ \prod_{i=1}^q \rho_{n,i}(Z) \right\} < \infty, \quad n \geq 1. \quad (21)$$

If ν is σ -finite and satisfies (19), then $\Pi_\nu \sim \text{PRM}(\nu)$ has finitely many particles in every product window on one probability-one event. Indeed, the counts are Poisson with finite means, first simultaneously for the countable family with rational thresholds, and then for all thresholds by inclusion. For $q \geq 2$ this locality does not imply finiteness on sets that constrain only one component.

To this end, we establish the existence of a stochastic representer for a given measure ν .

Lemma 2.3. (i) *If $\nu \in M_\sigma(\mathcal{X})$, there exist a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and an $(\mathcal{X}, \mathcal{E}_{\text{cyl}})$ -valued random element Z such that $\nu = \nu_Z$.*
(ii) *If $\nu \in M_\sigma(\mathcal{C})$ for a regular cylindrical path carrier $\mathcal{C}$, there exists a $\mathcal{C}$ -valued random element Z with $\nu = \nu_Z$ and the field $(t, \omega) \mapsto Z(\omega)(t)$ is jointly measurable.*

3 Stationarity of mPps and representer's continuity

For the structural results let $E = \mathcal{X}$ or $E = \mathcal{C}$ with respective σ -field $\mathcal{E}_E = \mathcal{E}_{\text{cyl}}$ or $\mathcal{E}_{\mathcal{C}}$. For an E -valued random element Z shifts commute with scaling and hence

$$\nu_{B^a Z} = (B^a)_\# \nu_Z = \nu_Z \circ (B^a)^{-1}, \quad a \in \mathbb{T}. \quad (22)$$

We write shift invariance as

$$\nu_Z \circ (B^a)^{-1} = \nu_Z, \quad a \in \mathbb{T}. \quad (23)$$

Let $\mathfrak{H}(E)$ consist of the non-negative measurable functionals $F : E \rightarrow [0, \infty]$ satisfying

$$F(S_r f) = \chi(r) F(f), \quad r \in G, \quad f \in E. \quad (24)$$

For scalar stationarity criteria see [8, 9, 13].

Theorem 3.1. *Let Z be an E -valued random element with $\nu = \nu_Z \in M_\sigma(E)$ and let $\Pi_\nu \sim \text{PRM}(\nu)$. For each $a \in \mathbb{T}$ the following conditions are equivalent:*

$$(i) \quad (B^a)_\# \Pi_\nu \stackrel{d}{=} \Pi_\nu.$$

$$(ii) \quad (B^a)_\# \nu = \nu.$$

$$(iii) \quad \text{For every } F \in \mathfrak{H}(E)$$

$$\mathbb{E}\{F(B^a Z)\} = \mathbb{E}\{F(Z)\}. \quad (25)$$

The equivalence of (i) and (ii) holds for any sigma-finite measure ν on $(E, \mathcal{E}_E)$. Moreover for every E -valued random element W

$$\nu_W = \nu_Z \iff \mathbb{E}\{F(W)\} = \mathbb{E}\{F(Z)\}, \quad F \in \mathfrak{H}(E). \quad (26)$$

Assume next that

$$\mathbb{P}\{\mathfrak{W}_Z(e) > 0\} = 1, \quad p_e := \mathbb{E}\{\mathfrak{W}_Z(e)\} \in (0, \infty) \quad (27)$$

and define the law of an E -valued random element Θ (for a random element U its law is abbreviated as $\mathcal{L}(U)$) by

$$\mathbb{E}\{\Gamma(\Theta)\} = \frac{1}{p_e} \mathbb{E}\left\{\mathfrak{W}_Z(e) \Gamma\left(\frac{Z}{|Z(e)|}\right)\right\}, \quad \Gamma \geq 0 \text{ measurable on } E, \quad (28)$$

where absolute values and quotients are componentwise (so $|Z(e)| = (|Z_1(e)|, \dots, |Z_q(e)|)$). Clearly, $|\Theta(e)| = \mathbf{1}$ almost surely; for $q = 1$ this is the local process, see [8, 13].

Corollary 3.2. *In the setting of Theorem 3.1 if (27) holds, then $\nu_Z = p_e \nu_\Theta$ and $(p_e, \mathcal{L}(\Theta))$ depends only on ν_Z . Moreover Π_ν is stationary iff for every $h \in \mathbb{T}$ and non-negative measurable H on E (set the whole term to zero when $\mathfrak{W}_\Theta(h^{-1}) = 0$)*

$$\mathbb{E}\left\{\mathfrak{W}_\Theta(h^{-1})H\left(\frac{B^h\Theta}{|\Theta(h^{-1})|}\right)\right\} = \mathbb{E}\{H(\Theta)\}. \quad (29)$$

For positive paths on $(\mathbb{R}, +)$, compare (29) with [1, Thm. 4.1], with vector parameter $\lambda = \mathbf{1}$ under the logarithmic correspondence.

Next we apply the above criteria to the first two examples in the Introduction.

Proposition 3.3. *Let X be an $\mathbb{R}^q$ -valued random field with $X(e) = 0$ and shift-invariant ν_{eX} . For $A \in \mathbb{R}^{q \times q}$ setting $\tilde{Z} = e^{AX}$ and $\mathcal{G}_A = \sigma\{AX(t) : t \in \mathbb{T}\}$ we have that $\nu_{\tilde{Z}}$ is shift-invariant iff*

$$\mathbb{E}\left\{e^{\langle \mathbf{1}, X(h) \rangle} \mid \mathcal{G}_A\right\} = e^{\langle \mathbf{1}, AX(h) \rangle} \quad \text{almost surely for every fixed } h \in \mathbb{T}. \quad (30)$$

If A is invertible, this is equivalent to

$$\langle A^\top \mathbf{1} - \mathbf{1}, X(h) \rangle = 0 \quad \text{almost surely for every fixed } h. \quad (31)$$

The matrix condition need not be necessary for singular A : $A = \text{diag}(1, 0)$ applied to two independent positive scalar representers with unit anchors and shift-invariant intensities gives $(Z_1, 1)$, whose intensity is shift-invariant.

Example 3.4. Under the assumptions of Proposition 3.3 assuming $A^\top \mathbf{1} = \mathbf{1}$ we have with $Z = e^X$

$$\tilde{Z}_i(t) = \prod_{j=1}^q Z_j(t)^{a_{ij}}, \quad \prod_{i=1}^q \tilde{Z}_i(t) = \prod_{j=1}^q Z_j(t)$$

and $\nu_{\tilde{Z}}$ is shift-invariant. The pointwise map $\Phi_A(x) = e^{A \log x}$ on $(0, \infty)^q$, with any Borel extension to $\mathbb{R}^q$, induces a cylinder-measurable map on $\mathcal{X}$. On positive paths it preserves the coordinate product and commutes with shifts and reanchoring, so the assertion also follows from (29) with $H \circ \Phi_A$.

If X has continuous paths, or càdlàg paths on $(\mathbb{R}, +)$, then so does AX . If further A is entrywise nonnegative and ν_Z satisfies (19), then so does $\nu_{\tilde{Z}}$. Indeed, for compact $K_1, \dots, K_q$ and $K = \bigcup_{i=1}^q K_i$

$$\prod_{i=1}^q \sup_{t \in K_i} \tilde{Z}_i(t) \leq \prod_{j=1}^q \left(\sup_{t \in K} Z_j(t) \right)^{\sum_{i=1}^q a_{ij}} = \prod_{j=1}^q \sup_{t \in K} Z_j(t).$$

For $q = 2$ on $(\mathbb{R}, +)$, take

$$A = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

Let N_1, N_2 be independent Poisson processes with rates $\lambda_1, \lambda_2 > 0$, and set $L_j^+(t) = N_j(t) - \lambda_j(e-1)t$, $j = 1, 2$. Complete the scalar marks independently using Proposition 3.5. Their matrix transform has dependent non-Gaussian logarithmic components: for $t > 0$

$$\text{Cov}(\log \tilde{Z}_1(t), \log \tilde{Z}_2(t)) = \frac{3t}{16}(\lambda_1 + \lambda_2) > 0.$$

On $(\mathbb{R}, +)$, call $Z = e^X$ a *log-Lévy mark* if

$$X(t) = \begin{cases} L^+(t), & t \geq 0, \\ -L^-((-t)-), & t < 0, \end{cases} \quad (32)$$

where L^+ and L^- are independent $\mathbb{R}^q$ -valued Lévy processes starting at zero.

Proposition 3.5. *For a log-Lévy mark Z*

$$\nu_Z \text{ is shift-invariant} \iff \mathbb{P}\{L^-(1) \in dx\} = e^{\langle \mathbf{1}, x \rangle} \mathbb{P}\{L^+(1) \in dx\}. \quad (33)$$

The proposition gives the following construction. For any vector Lévy process L with cumulant $\psi(u) = \log \mathbb{E}\{e^{\langle u, L(1) \rangle}\}$ and $\psi(\mathbf{1}) < \infty$, choose b with $\langle \mathbf{1}, b \rangle = \psi(\mathbf{1})$, set $L^+(t) = L(t) - bt$ and let L^- be an independent Lévy process with

$$\mathbb{P}\{L^-(t) \in dx\} = e^{\langle \mathbf{1}, x \rangle} \mathbb{P}\{L^+(t) \in dx\}.$$

The choice of b makes the right-hand side a probability convolution semigroup, and (33) holds. For the scalar construction and duality, see [14, Sec. 1.2 and Thm. 1.2]. If (33) holds, applying it to AL^+ and AL^- reduces (30) to

$$\mathbb{E}\left\{e^{\langle \mathbf{1}, Y \rangle} \mid AY\right\} = e^{\langle \mathbf{1}, AY \rangle}, \quad Y = L^+(1),$$

which remains valid for singular A .

Example 3.6. Take $q = 2$ and independent Poisson processes N_0, N_1, N_2 with rates $\lambda_0, \lambda_1, \lambda_2 > 0$. In Proposition 3.5, set

$$L_i(t) = N_0(t) + N_i(t), \quad b_i = \lambda_i(e - 1) + \frac{\lambda_0}{2}(e^2 - 1), \quad i = 1, 2.$$

The common process N_0 creates dependence. For the negative half, take independent Poisson processes $\widehat{N}_0, \widehat{N}_1, \widehat{N}_2$, independent also of N_0, N_1, N_2 , with rates $\lambda_0 e^2, \lambda_1 e, \lambda_2 e$, and put

$$X_i(-t) = -\widehat{N}_0(t-) - \widehat{N}_i(t-) + b_i t, \quad t > 0.$$

It follows easily that (21) and (46) hold.

3.1 Stochastic continuity of the representer

For continuity we additionally require joint measurability of $(t, \omega) \mapsto Z(t, \omega)$ on $(\mathbb{T} \times \Omega, \mathcal{B}(\mathbb{T}) \otimes \mathcal{F})$. We keep the cylindrical represented measure and shift-invariance condition (23).

Theorem 3.7. *Let Z be a jointly measurable $\mathbb{R}^q$ -valued random field on $\mathbb{T}$. If its cylindrical represented measure ν_Z is σ -finite and satisfies (23), then Z is stochastically continuous.*

Conditions (12) and (13) suffice for σ -finiteness.

For a $\mathcal{C}$ -valued representer, joint evaluation gives joint field measurability, and inclusion $\iota : \mathcal{C} \rightarrow \mathcal{X}$ preserves shifts and σ -finiteness. Indeed, a finite-measure trace cover $A_n = \mathcal{C} \cap \widetilde{A}_n$ lifts to the cylindrical cover $(\widetilde{A}_n)$ together with the $\iota_{\#}\nu_Z$ -null set $\mathcal{X} \setminus \bigcup_{n \geq 1} \widetilde{A}_n$. The theorem therefore applies when the carrier measure is sigma-finite and satisfies (23).

For $q = 1$, joint measurability, (23) and $\mathbb{E}\{|Z(e)|\} < \infty$ already imply L^1 -continuity, without σ -finiteness. Radial integration identifies $\mathbb{E}\{|Z(s) - Z(t)|\}$ with $\nu_Z\{f : |f(s) - f(t)| > 1\}$ and makes $\mathbb{E}\{|Z(t)|\}$ constant. Thus the increment L^1 -distance is translation invariant and Lemma 7.2 applies. For related scalar results, see e.g., [6, Lem. 2], [7, Thm. 3.1] and [15, Thm. A.1 and Cor. A.3].

Example 3.8. Let $\mathsf{T} = (\mathbb{R}, +)$, $q \geq 2$ and take $\mathcal{C} = C(\mathbb{R}, \mathbb{R}^q)$. Let W be a continuous stationary Gaussian field with $\text{Var}(W_i(0)) > 0$ for $i = 1, \dots, q$. Choose $c \in \mathbb{R}^q \setminus \{0\}$ with $\langle \mathbf{1}, c \rangle = 0$ and put $Z_i(t) = \exp(c_i t) W_i(t)$. Then Z is Gaussian and

$$B^h Z \stackrel{d}{=} S_{(\exp(-hc_i))_{i=1}^q} Z, \quad \nu_{B^h Z} = \exp(-h \langle \mathbf{1}, c \rangle) \nu_Z = \nu_Z.$$

Radial integration and Fernique's theorem imply $\nu_Z \in M_\sigma(\mathcal{C})$ and (19). Thus Π_{ν_Z} is stationary by Theorem 3.1, although Z is not: $\text{Var}(Z_i(t)) = \exp(2c_i t) \text{Var}(W_i(0))$ varies with t whenever $c_i \neq 0$.

4 Finite mixtures

For continuous marks we use $\mathcal{C} = C(\mathsf{T}, \mathbb{R}^q)$ with its compact-open Borel sigma-field, which equals $\mathcal{E}_{\mathcal{C}}$. When log-Lévy marks are included, we take $\mathsf{T} = (\mathbb{R}, +)$ and $\mathcal{C} = D(\mathbb{R}, \mathbb{R}^q)$ with its trace cylindrical σ -field. Logarithms and quotients of positive paths are taken componentwise. A continuous mark is called log-Gaussian if it has strictly positive paths and its logarithm is a Gaussian field. On $\mathbb{R}$ we also use log-Lévy marks $Z = \exp(X)$ with X as in (32), without assuming (33).

After merging equal laws, let $Z_1, \dots, Z_n$ be strictly positive $\mathcal{C}$ -valued random elements with distinct laws. Fix $p_k > 0$ with $\sum_{k=1}^n p_k = 1$, and let Z have the finite mixture law

$$\mathcal{L}(Z) = \sum_{k=1}^n p_k \mathcal{L}(Z_k), \quad \nu_Z = \sum_{k=1}^n p_k \nu_{Z_k}. \quad (34)$$

Theorem 4.1. *Suppose each Z_k is log-Gaussian or log-Lévy, the latter case requiring $\mathsf{T} = \mathbb{R}$ and $Z_k(e) = \mathbf{1}$ almost surely. If T is connected, then*

$$\Pi_{\nu_Z} \text{ is stationary} \iff \Pi_{\nu_{Z_k}} \text{ is stationary for } k = 1, \dots, n. \quad (35)$$

If all marks are log-Gaussian, then ν_Z satisfies (19). If also $Z_k(e) = \mathbf{1}$, $k = 1, \dots, n$, almost surely, then on an arbitrary T stationarity is equivalent to the existence of a unique continuous homomorphism $\varpi : \mathsf{T} \rightarrow \mathfrak{S}_n$, with $\varpi_h = \varpi(h)$ and $\mathfrak{S}_n$ discrete, such that

$$p_k \mathbb{E} \left\{ \mathfrak{W}_{Z_k}(h^{-1}) H \left(\frac{B^h Z_k}{Z_k(h^{-1})} \right) \right\} = p_{\varpi_h(k)} \mathbb{E} \{ H(Z_{\varpi_h(k)}) \}, \quad k = 1, \dots, n, \quad h \in \mathsf{T}, \quad (36)$$

for every non-negative measurable H on $\mathcal{C}$.

Condition (19) is not automatic for log-Lévy marks. For example, take $q = 2$ in (32), with L^+, L^- independent copies of $(J, -J)$, where J is an increasing compound Poisson process with $\mathbb{E}\{e^{J(1)}\} = \infty$. Since the coordinate sum vanishes, (33) holds and Proposition 3.5 implies shift-invariance of ν_Z , but

$$\mathbb{E} \left\{ \sup_{t \in [0,1]} Z_1(t) \sup_{t \in [0,1]} Z_2(t) \right\} = \mathbb{E}\{e^{J(1)}\} = \infty.$$

Single Gaussian marks. For a single log-Gaussian mark Z with $Z(e) = \mathbf{1}$, put $X = \log Z$, $m(t) = \mathbb{E}\{X(t)\}$, and $\Sigma(s, t)_{ij} = \text{Cov}\{X_i(s), X_j(t)\}$, $i, j = 1, \dots, q$. With $b(t) = m(t) + \frac{1}{2}\Sigma(t, t)\mathbf{1}$, the Gaussian case $n = 1$ of Theorem 4.1 says that Π_{ν_Z} is stationary iff $\langle b(t), \mathbf{1} \rangle = 0$ and

$$\Sigma(hs, ht) - \Sigma(hs, h) - \Sigma(h, ht) + \Sigma(h, h) = \Sigma(s, t), \quad (37)$$

$$b(t) - b(s) + \frac{1}{2}\{\Sigma(t, s) - \Sigma(s, t)\}\mathbf{1} = b(s^{-1}t), \quad (38)$$

for all $h, s, t \in \mathbb{T}$. Indeed, the Gaussian transfer at s^{-1} has mass $\exp\{\langle \mathbf{1}, b(s) \rangle\}$; its covariance equality is (37), and its mean equality is

$$m(s^{-1}t) = m(t) - m(s) + (\Sigma(t, s) - \Sigma(s, s))\mathbf{1}.$$

By (37), $\Sigma(s^{-1}t, s^{-1}t) = \Sigma(t, t) + \Sigma(s, s) - \Sigma(t, s) - \Sigma(s, t)$, so this mean equality is equivalent to (38). For $q \geq 3$, (37) is not determined by the matrix-variogram conditions in [1, Thm. 4.4].

Example 4.2. Connectedness cannot be omitted in general. Let $\mathbb{T} = \mathbb{Z}/2\mathbb{Z}$ and $q = 1$. Let Z_1, Z_2 be log-Gaussian marks with $Z_k(0) = 1, k = 1, 2$. If $\log Z_1(1)$ and $\log Z_2(1)$ have distributions $N(0, 2)$ and $N(-2, 2)$, respectively, then for every non-negative measurable H ,

$$\mathbb{E}\left\{Z_1(1)H\left(\frac{B^1 Z_1}{Z_1(1)}\right)\right\} = e \mathbb{E}\{H(Z_2)\}, \quad \mathbb{E}\left\{Z_2(1)H\left(\frac{B^1 Z_2}{Z_2(1)}\right)\right\} = e^{-1} \mathbb{E}\{H(Z_1)\}.$$

Consequently, the mixture $\mathcal{L}(Z) = \{\mathcal{L}(Z_1) + e\mathcal{L}(Z_2)\}/(1 + e)$ generates a stationary particle system, although neither component system is stationary.

Example 4.3. The componentwise conclusion can fail outside the two classes in Theorem 4.1, even for two bounded marks on $\mathbb{T} = \mathbb{R}$. Indeed, let $q = 1$, set

$$z_\phi(t) = \frac{2 + \cos(t + \phi)}{2 + \cos \phi},$$

and let Φ have density $(2 + \cos \phi)/(4\pi)$ on $[0, 2\pi)$. Set $Z = z_\Phi$, and let Z_1, Z_2 have the conditional laws of Z given $\Phi \in [0, \pi)$ and $\Phi \in [\pi, 2\pi)$, respectively. Both conditioning events have probability $1/2$, so $\mathcal{L}(Z) = \{\mathcal{L}(Z_1) + \mathcal{L}(Z_2)\}/2$.

With phases read modulo 2π , we have

$$\frac{B^h z_\phi}{z_\phi(-h)} = z_{\phi-h}.$$

Consequently, for every non-negative measurable H

$$\mathbb{E}\left\{Z(-h)H\left(\frac{B^h Z}{Z(-h)}\right)\right\} = \frac{1}{4\pi} \int_0^{2\pi} (2 + \cos(\phi - h))H(z_{\phi-h}) d\phi = \mathbb{E}\{H(Z)\}$$

implying ν_Z is shift-invariant by (29). However direct integration gives

$$\mathbb{E}\{Z_1(t)\} = 1 - \frac{\sin t}{\pi}, \quad \mathbb{E}\{Z_2(t)\} = 1 + \frac{\sin t}{\pi}.$$

Since $Z_i(0) = 1$, neither component satisfies (29) with $H = 1$.

All paths are smooth and satisfy $1/3 \leq Z_i(t) \leq 3$, so (21) holds. The bounded, nonconstant $\log Z_i(\pi)$ shows that neither component is log-Gaussian.

Example 4.4. The finiteness assumption in the Gaussian mixture result cannot be omitted, even for $\mathbb{T} = (\mathbb{R}, +)$. Let $q = 1$, let $X \sim N(0, 1)$, and let Φ be uniform on $[0, 2\pi)$ and independent of X . For $\theta \in [0, 2\pi)$ define

$$a_\theta(t) = \cos(\theta + t) - \cos \theta, \quad Z_\theta(t) = \exp\{a_\theta(t)X - \frac{1}{2}a_\theta(t)^2\}, \quad \bar{Z} = Z_\Phi.$$

Conditionally on $\Phi = \theta$, this is a log-Gaussian mark with $Z_\theta(0) = 1$. Gaussian tilting of the logarithm and normalisation at $-h$, $h \in \mathbb{R}$, give

$$\mathbb{E}\{Z_\theta(-h)\} = 1, \quad \mathcal{L}_{\mathbb{P}_{\theta,h}}\left(\frac{B^h Z_\theta}{Z_\theta(-h)}\right) = \mathcal{L}(Z_{\theta-h}), \quad (39)$$

where $d\mathbb{P}_{\theta,h} = Z_\theta(-h) d\mathbb{P}$ and the phase is read modulo 2π . Indeed, under $\mathbb{P}_{\theta,h}$ the variable $X - a_\theta(-h)$ is standard Gaussian, and

$$\log \frac{Z_\theta(t-h)}{Z_\theta(-h)} = a_{\theta-h}(t)\{X - a_\theta(-h)\} - \frac{1}{2}a_{\theta-h}(t)^2.$$

Consequently, for every non-negative measurable $H : \mathcal{C} \rightarrow [0, \infty]$

$$\mathbb{E}\left\{\bar{Z}(-h)H\left(\frac{B^h \bar{Z}}{\bar{Z}(-h)}\right)\right\} = \frac{1}{2\pi} \int_0^{2\pi} \mathbb{E}\{H(Z_{\theta-h})\} d\theta = \mathbb{E}\{H(\bar{Z})\}.$$

Thus (29) shows that $\nu_{\bar{Z}}$ is shift-invariant.

No fixed-phase component is stationary. If ν_{Z_θ} were shift-invariant, (29) and (39) would give $\mathcal{L}(Z_{\theta-h}) = \mathcal{L}(Z_\theta)$ for every h . This contradicts

$$\text{Var}\{\log Z_{\theta-h}(\pi)\} = 4 \cos^2(\theta - h).$$

Thus every conditional particle system $\Pi_{\nu_{Z_\theta}}$ is nonstationary.

Moreover, $\log \bar{Z}$ is not a finite mixture of Gaussian laws. Indeed, put $L = \log \bar{Z}(\pi)$ and $v_\theta = 4 \cos^2 \theta$. Conditionally on $\Phi = \theta$, the variable L has distribution $N(-v_\theta/2, v_\theta)$, so

$$M(r) := \mathbb{E}\{e^{rL}\} = \mathbb{E}\{e^{(r^2-r)v_\Phi/2}\}.$$

Since $\text{ess sup } v_\Phi = 4$ and $\mathbb{P}\{v_\Phi = 4\} = 0$, the essential-supremum property and dominated convergence give

$$\lim_{|r| \rightarrow \infty} \frac{2 \log M(r)}{r^2} = 4, \\ e^{-2(r^2-r)} M(r) = \mathbb{E}\{e^{-(4-v_\Phi)(r^2-r)/2}\} \longrightarrow 0, \quad r \rightarrow \pm\infty.$$

If L were a finite Gaussian mixture, the first limit would force a component of variance 4, with some mean μ and weight $p > 0$. Its contribution $pe^{(\mu+2)r}$ to $e^{-2(r^2-r)} M(r)$ cannot tend to zero at both ends, a contradiction.

More strongly, no finite mixture of log-Gaussian marks generates $\nu_{\bar{Z}}$, even if the component marks are not normalised at zero. Otherwise, Corollary 3.2 would identify the canonical spectral law of the common intensity with $\mathcal{L}(\bar{Z})$. For each log-Gaussian component, the tilt and normalisation in (28) preserve Gaussianity of the logarithm. The same canonical law would therefore be a finite mixture of log-Gaussian laws, contradicting the preceding paragraph.

Finally, $\bar{Z}(t) \leq e^{2|X|}$ and $\mathbb{E}\{e^{2|X|}\} < \infty$ give (21) for $\bar{Z}$.

Example 4.5. Example 4.3 also yields a matrix counterexample with independent nondegenerate input coordinates. Let U_1, U_2 have its two component laws, and independently let

$$W(t) = \exp\{\sqrt{2}B(t) - |t|\}, \quad t \in \mathbb{R},$$

where B is standard two-sided Brownian motion. Then ν_W is shift-invariant by (37)–(38), and $\mathbb{E}\{W(t)\} = 1$. For $k = 1, 2$, put

$$V_k = (U_k, W), \quad \tilde{V}_k = e^{A \log V_k}, \quad A = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

Independence and (29) give shift-invariance for the intensity of the equally weighted mixture of V_1, V_2 . By Example 3.4, the same holds for the mixture of $\tilde{V}_1, \tilde{V}_2$, but product preservation gives

$$\mathbb{E}\{\tilde{V}_{1,1}(t)\tilde{V}_{1,2}(t)\} = 1 - \frac{\sin t}{\pi}, \quad \mathbb{E}\{\tilde{V}_{2,1}(t)\tilde{V}_{2,2}(t)\} = 1 + \frac{\sin t}{\pi}.$$

Neither transformed component satisfies (29) with $H = 1$. All paths are continuous, and for compact K_1, K_2 and $K = K_1 \cup K_2$,

$$\prod_{i=1}^2 \sup_{t \in K_i} \tilde{V}_{k,i}(t) \leq 3 \sup_{t \in K} W(t),$$

whose expectation is finite and hence the intensities satisfy (19).

5 Stationary systems from product clusters

Throughout this section, $\mathbb{T}$ is an lcscH group with identity e and left Haar measure λ , and $\mathcal{C}$ is a regular cylindrical path carrier over $\mathbb{T}$. Further Z is a $\mathcal{C}$ -valued random element and for $f \in \mathcal{C}$ define

$$\mathcal{S}_\Pi(f) = \int_{\mathbb{T}} \mathfrak{W}_f(t) \lambda(dt). \quad (40)$$

The map $\mathcal{S}_\Pi$ is $\mathcal{E}_\mathcal{C}$ -measurable by joint evaluation and Tonelli's theorem. Moreover, we have

$$\mathcal{S}_\Pi(S_r f) = \chi(r) \mathcal{S}_\Pi(f), \quad \mathcal{S}_\Pi(B^a f) = \mathcal{S}_\Pi(f).$$

Definition 5.1. A $\mathcal{C}$ -valued random element Q is a product cluster field if it satisfies (17) and, for every tuple $\mathbf{K} = (K_1, \dots, K_q)$ of nonempty compact subsets of $\mathbb{T}$

$$\Lambda_{\mathbf{K}}(Q) := \int_{\mathbb{T}} \mathbb{E} \left\{ \prod_{i=1}^q \rho_{K_i, i}(B^v Q) \right\} \lambda(dv) < \infty. \quad (41)$$

By monotonicity and cofinality of the increasing compact exhaustion, it is enough in (41) to check $\mathbf{K} = (K_n, \dots, K_n)$, $n \geq 1$.

For any $\mathcal{C}$ -valued random element Q , the formula

$$\nu_Q^{\text{cl}}(A) := \int_{\mathbb{T}} \mathbb{E} \left\{ \int_G \mathbb{I}_{\{S_r B^v Q \in A\}} \vartheta(dr) \right\} \lambda(dv), \quad A \in \mathcal{E}_\mathcal{C} \quad (42)$$

defines an extended-valued measure on $\mathcal{C}$.

If Q is a product cluster field, let

$$\tilde{\Pi}_Q = \sum_{j \geq 1} \delta_{(V_j, R_j, Q_j)} \sim \text{PRM}(\lambda \otimes \vartheta \otimes \mathcal{L}(Q))$$

and map its atoms to trajectories by

$$\Pi_Q^{\text{cl}} = \sum_{j \geq 1} \delta_{S_{R_j} B^{V_j} Q_j}. \quad (43)$$

Lemma 5.2. *If Q is a product cluster field, then Π_Q^{cl} is stationary, $\nu_Q^{\text{cl}} \in M_\sigma(\mathcal{C})$ and this intensity satisfies (19) with*

$$\nu_Q^{\text{cl}}(U_{\mathbf{n}, \varepsilon}) = \left(\prod_{i=1}^q \varepsilon_i^{-1} \right) \Lambda_{(K_{n_1}, \dots, K_{n_q})}(Q). \quad (44)$$

Remark 5.3. Let the random variable N be $\mathbb{T}$ -valued independent of the product cluster field Q , with a strictly positive density p_N with respect to λ and define

$$Z^{Q, N} = p_N(N)^{-1/q} B^N Q. \quad (45)$$

Then $\nu_{Z^{Q, N}} = \nu_Q^{\text{cl}}$: the scaling vector in (45) has character $p_N(N)^{-1}$, which cancels the density of N in the radial integral. For the scalar random-shift construction, see [10, Def. 1.2, Eq. (1.8), and Thm. 3.4].

Proposition 5.4. *Let Z be a $\mathcal{C}$ -valued random element, and suppose that $\nu = \nu_Z$ is shift-invariant and satisfies (19). If*

$$\mathbb{P}\{\mathcal{S}_\Pi(Z) \in (0, \infty)\} = 1, \quad (46)$$

then $0 < p := \mathbb{E}\{\mathfrak{W}_Z(e)\} < \infty$. Define $\mathbb{P}^$ by $\frac{d\mathbb{P}^*}{d\mathbb{P}} = \frac{\mathfrak{W}_Z(e)}{p}$. Under $\mathbb{P}^*$, with expectation $\mathbb{E}^*$, define*

$$Q^{\text{com}} = \left(\frac{p}{\mathcal{S}_\Pi(Z)} \right)^{1/q} Z. \quad (47)$$

Set $Q^{\text{com}} = Z$ where $\mathcal{S}_\Pi(Z) \notin (0, \infty)$. Then Q^{com} is a product cluster field under $\mathbb{P}^$, $\mathcal{S}_\Pi(Q^{\text{com}}) = p$ $\mathbb{P}^*$ -almost surely, and*

$$\nu = \nu_{Q^{\text{com}}}^{\text{cl}}. \quad (48)$$

The law of Q^{com} may depend on the representer. For $c \in \ker \chi$, replacing Z by $S_c Z$ leaves ν and $\mathbb{P}^*$ unchanged and replaces Q^{com} by $S_c Q^{\text{com}}$.

Remark 5.5. Proposition 5.4 assumes (46). For $q \geq 2$, condition (17) may hold even when $\mathbb{E}\{\mathfrak{W}_Z(h)\} = 0$ for every h . The hypothesis (46) is intrinsic to the represented intensity: the set $\{f : \mathcal{S}_\Pi(f) \notin (0, \infty)\}$ is scale invariant, so (11) makes (46) equivalent to this set having zero ν -mass.

Related scalar moving-maximum profiles appear in [16, eq. (1.4)].

Example 5.6. Take $\mathbb{T} = (\mathbb{R}, +)$ with $\lambda = \text{Leb}_1$, $q = 2$. Further set $\mathcal{C} = C(\mathbb{R}, [0, \infty)^2)$ and

$$Q_1(t) = Q_2(t) = (1 + t^2)^{-1/2}, \quad t \in \mathbb{R}.$$

Each component has infinite integral, whereas $\mathcal{S}_\Pi(Q) = \pi$. If $K_1, K_2 \subset [-B, B]$, then

$$\prod_{i=1}^2 \rho_{K_i, i}(B^v Q) \leq \frac{1}{1 + ((|v| - B)_+)^2}, \quad v \in \mathbb{R}.$$

This integrable bound makes Q a product cluster field. Hence $\nu := \nu_Q^{\text{cl}}/\pi$ is shift-invariant, belongs to $M_\sigma(\mathcal{C})$ and satisfies (19).

For a standard Cauchy variable N , define next

$$Z_i(t) = \sqrt{\frac{1 + N^2}{1 + (t - N)^2}}, \quad i = 1, 2.$$

The scaling character $1 + N^2$ cancels the same factor in the denominator of the density of N , so $\nu_Z = \nu$. Moreover, $Z(0) = \mathbf{1}$ and $\mathcal{S}_\Pi(Z) = \pi(1 + N^2) \in (0, \infty)$ almost surely. For each fixed $t \neq 0$, $\log Z_i(t)$ is bounded and nonconstant, so this normalized representer is not log-Gaussian. Radial integration also gives the explicit hit intensity

$$\nu\{f : \exists t \in [0, S], f(t) > \mathbf{1}\} = \frac{1}{\pi} \int_{\mathbb{R}} \frac{dv}{1 + \text{dist}(v, [0, S])^2} = 1 + \frac{S}{\pi}, \quad S \geq 0. \quad (49)$$

Thus the maximal index introduced below is $\mathcal{H}^\nu = 1/\pi$.

6 The maximal index on amenable unimodular groups

Throughout this section $\mathbb{T}$ is an amenable unimodular lscH group with identity e and Haar measure λ , and $\mathcal{C} = C(\mathbb{T}, [0, \infty)^q)$ has its compact-open Borel sigma-field. Choose nonempty compact windows $F_n, n \geq 1$ with $\lambda(F_n) > 0$ such that for every compact $C \subset \mathbb{T}$ containing e

$$\lim_{n \rightarrow \infty} \frac{\lambda(F_n C)}{\lambda(F_n)} = 1. \quad (50)$$

Such sequences exist in this setting, see [17, Remarks 5.6–5.7]. For compact $\mathbb{T}$ one may take $F_n = \mathbb{T}$. Unimodularity makes both right translations and inversion preserve λ .

For $f \in \mathcal{C}$, $y \in (0, \infty)^q$ and nonempty compact $K \subset \mathbb{T}$ put

$$E_y(f) = \{t \in \mathbb{T} : f(t) > y\}, \quad \mathcal{R}_{\text{sup}, K}(f) = \int_{(0, \infty)^q} \mathbb{I}_{\{E_y(f) \cap K \neq \emptyset\}} dy$$

and set further

$$\mathcal{R}_{\text{sup}}(f) = \int_{(0, \infty)^q} \mathbb{I}_{\{E_y(f) \neq \emptyset\}} dy, \quad \tau(f) = \lambda(E_1(f)). \quad (51)$$

Thus $\mathcal{R}_{\text{sup}}(f)$ is the volume of $\bigcup_{t \in \mathbb{T}} \prod_{i=1}^q (0, f_i(t))$; for $q = 1$ it is the usual supremum. Continuity, separability and Tonelli's theorem make these functionals measurable.

For a $\mathcal{C}$ -valued random element Z and $\nu = \nu_Z$ set

$$H^\nu(K) := \nu\{f : \exists t \in K, f(t) > \mathbf{1}\} = \mathbb{E}\{\mathcal{R}_{\text{sup}, K}(Z)\} = \int_{(0, \infty)^q} \mathbb{P}\{\exists t \in K : Z(t) > y\} dy. \quad (52)$$

Define the maximal index of ν with respect to $F_n, n \geq 1$ by

$$\mathcal{H}^\nu := \lim_{n \rightarrow \infty} \frac{H^\nu(F_n)}{\lambda(F_n)}$$

whenever the limit exists. Below Θ has the law of $Z/Z(e)$ under $\mathbb{P}^*$ and R is independent of Θ with independent 1-Pareto components satisfying $\mathbb{P}\{R_i > r\} = r^{-1}$ for $r \geq 1$.

Theorem 6.1. *Suppose that $\nu := \nu_Z$ is shift-invariant, (19) holds and Z satisfies (46). If $p, \mathbb{P}^*$ and Q^{com} are as in Proposition 5.4, then $\mathcal{H}^\nu$ exists, lies in $(0, \infty)$, and is independent of the chosen windows satisfying (50). It has the representations*

$$\mathcal{H}^\nu = \mathbb{E}^*\{\mathcal{R}_{\text{sup}}(Q^{\text{com}})\} = \mathbb{E}\left\{\mathfrak{W}_Z(e) \frac{\mathcal{R}_{\text{sup}}(Z)}{\mathcal{S}_\Pi(Z)}\right\}, \quad (53)$$

$$= \int_{\{f: f(e) > 1\}} \frac{\nu(df)}{\tau(f)} = p \mathbb{E}\left\{\frac{1}{\tau(S_R \Theta)}\right\}. \quad (54)$$

Here $0 \cdot \infty = 0$ and $1/\infty = 0$.

For $q = 1$ and $\mathbb{T} = \mathbb{Z}$ with counting Haar measure, $\mathcal{H}^\nu/p$ is the maximal index for exceedance at the origin in [8, Section 2.6 and Proposition 2.20].

Corollary 6.2. *Let $\mathbb{T} = (\mathbb{R}, +)$ with $\lambda = \text{Leb}_1$, and let Z be the log-Gaussian representer in (6), under the covariance assumption there, with $c := \mathbf{1}^\top V \mathbf{1} > 0$. Then the constant $\mathcal{H}$ in (7) exists and satisfies*

$$\mathcal{H} = \mathcal{H}^{\nu_Z} = \mathbb{E}\left\{\frac{\mathcal{R}_{\text{sup}}(Z)}{\mathcal{S}_\Pi(Z)}\right\} = \mathbb{E}\left\{\frac{1}{\tau(S_R Z)}\right\} \in (0, \infty), \quad (55)$$

where R is as in Theorem 6.1 and independent of Z . In particular,

$$\tau(S_R Z) = \text{Leb}_1\{t \in \mathbb{R} : Y_{\kappa, V}(t) - S_{\kappa, V}(t)\mathbf{1} + \log R > 0\} \in (0, \infty) \quad \text{almost surely.}$$

7 Further results and proofs

Lemma 7.1. *Let $\mathcal{C}$ be a regular cylindrical path carrier and let $\nu \in M(\mathcal{C})$.*

(i) *If ν is σ -finite, then*

$$\nu\{f \in \mathcal{C} : f_i \equiv 0\} = 0, \quad i = 1, \dots, q. \quad (56)$$

(ii) *If (19) and (56) hold, then ν is σ -finite.*

For a $\mathcal{C}$ -valued Z , σ -finiteness of ν_Z implies (17); the converse holds under (19).

Proof of Lemma 7.1. The sets $F_i = \{f \in \mathcal{C} : f_i \equiv 0\}$ are measurable by (16), and part (i) follows from Lemma 2.1 on $\mathcal{C}$. Conversely, the windows $U_{(n, \dots, n), (1/n, \dots, 1/n)}$, $n \geq 1$, cover $\mathcal{C} \setminus \bigcup_{i=1}^q F_i$ and have finite mass under (19). Adding the null set $\bigcup_{i=1}^q F_i$ proves σ -finiteness. Finally, (11) applied to F_i gives $\nu_Z(F_i) = 0$ exactly when $\mathbb{P}\{Z_i \equiv 0\} = 0$. $\square$

Lemma 7.2. *Let H be an lcscH group with left Haar measure λ_H , and let $U : H \times \Omega \rightarrow \mathbb{R}$ be jointly measurable. If*

$$\int_K \mathbb{E}\{|U(t)|\} \lambda_H(dt) < \infty \quad \text{for every compact } K \subset H \quad (57)$$

and

$$\mathbb{E}\{|U(as) - U(at)|\} = \mathbb{E}\{|U(s) - U(t)|\}, \quad a, s, t \in H \quad (58)$$

hold, then

$$\mathbb{E}\{|U(s) - U(t)|\} \longrightarrow 0 \quad \text{as } s \rightarrow t. \quad (59)$$

Proof of Lemma 7.2. Put $d_U(h) = \mathbb{E}\{|U(e_H) - U(h)|\}$, so (58) gives $\mathbb{E}\{|U(s) - U(t)|\} = d_U(s^{-1}t)$. Right translations $R_h F(t, \omega) = F(th, \omega)$ are strongly continuous on $L^1(\lambda_H \otimes \mathbb{P})$. Indeed, with the modular function defined by

$$\int_H f(th) \lambda_H(dt) = \Delta_H(h)^{-1} \int_H f(t) \lambda_H(dt),$$

we have $\|R_h F\|_1 = \Delta_H(h)^{-1} \|F\|_1$. Continuity first follows for finite sums $\sum_{j=1}^m \varphi_j(t) \mathbb{I}_{A_j}(\omega)$ with $\varphi_j \in C_c(H)$, then for all $F \in L^1$ by density and continuity of Δ_H . Choose a compact K of positive Haar measure and a compact identity neighbourhood V . By (57), $F = \mathbb{I}_{KV} U \in L^1(\lambda_H \otimes \mathbb{P})$, and for $h \in V$

$$\lambda_H(K) d_U(h) = \int_K \mathbb{E}\{|F(t) - F(th)|\} \lambda_H(dt) \leq \|F - R_h F\|_1 \longrightarrow 0 \quad \text{as } h \rightarrow e_H.$$

Since $s^{-1}t \rightarrow e_H$ as $s \rightarrow t$, the claim follows. $\square$

Proof of Lemma 2.1. Choose a countable measurable cover (A_n) of E with $\nu(A_n) < \infty$. Every trace cylindrical set depends on countably many restricted evaluations, so all the A_n belong to $\sigma\{\pi_h|_E : h \in D\}$ for one nonempty countable D . For $i = 1, \dots, q$, put $N_i = \{f : f_i(h) = 0 \text{ for all } h \in D\}$. Scaling coordinate i leaves all evaluations on D unchanged on N_i . Hence $A_n \cap N_i$ is invariant under this scaling, and homogeneity gives

$$\nu(A_n \cap N_i) = \frac{1}{2} \nu(A_n \cap N_i) = 0.$$

Thus $\nu(N_i) = 0$ for $i = 1, \dots, q$ proving the first assertion. The set V_D^c is scale invariant. If $\nu = \nu_Z$, its zero mass and (11) imply $\mathbb{P}\{Z \in V_D^c\} = 0$. $\square$

Proof of Lemma 2.3. Let $E = \mathcal{X}$ or $E = \mathcal{C}$, with its respective σ -field $\mathcal{E}_E$. Use the countable set D supplied by Lemma 2.1. The evaluations $g_{h,i}(f) = |f_i(h)|$, $h \in D$, are measurable, satisfy $g_{h,i}(S_r f) = r_i g_{h,i}(f)$ and form countable detectors on the ν -conull set V_D . Thus [4, Thm. 2.2(iv) $\Rightarrow$ (i)], applied directly on $(E, \mathcal{E}_E)$, yields an E -valued Z with $\nu = \nu_Z$. For $E = \mathcal{C}$, Definition 2.2(a) implies joint measurability. $\square$

Proof of Theorem 3.1. In view of the mapping theorem

$$(B^a)_\# \Pi_\nu \sim \text{PRM}((B^a)_\# \nu)$$

and thus (ii) implies (i). Conversely, equal laws of Poisson random measures have equal mean measures, so (i) implies (ii). This argument uses only σ -finiteness of ν .

It remains to prove (26). For $\mathbf{h} \in \mathbb{T}^q$, put

$$\mathcal{A}_{\mathbf{h}}^{\circ} = \{f : 1 < |f_i(h_i)| \leq 2, i = 1, \dots, q\}, \quad E_{\mathbf{h}} = \{f : f_i(h_i) \neq 0, i = 1, \dots, q\}.$$

If Y is any E -valued random element, $F \in \mathfrak{H}(E)$ and $A \subset E_{\mathbf{h}}$ is measurable and invariant under all componentwise scalings, radial integration gives

$$\int_{\mathcal{A}_{\mathbf{h}}^{\circ}} F(f) \mathbb{I}_A(f) \nu_Y(df) = (\log 2)^q \mathbb{E}\{F(Y) \mathbb{I}_A(Y)\}. \quad (60)$$

Suppose that $\nu_W = \nu_Z$. Lemma 2.1 supplies one countable D for which the sectors $E_{\mathbf{h}}$, $\mathbf{h} \in D^q$, cover a conull set under both laws. Enumerate D^q and disjointify these invariant sectors. Equality of the measures on the left of (60), followed by summation over the disjoint pieces, gives $\mathbb{E}\{F(W)\} = \mathbb{E}\{F(Z)\}$ for every $F \in \mathfrak{H}(E)$. Conversely, for $A \in \mathcal{E}_E$ the orbit functional

$$F_A(f) = \int_G \mathbb{I}_{\{S_r f \in A\}} \vartheta(dr)$$

belongs to $\mathfrak{H}(E)$. The assumed moment equality applied to F_A gives the same expectation under W and Z , hence $\nu_W(A) = \nu_Z(A)$. This proves (26). Taking $W = B^a Z$ and using (22) proves the equivalence of (ii) and (iii). $\square$

Proof of Corollary 3.2. With

$$A_e = \{f \in E : |f_i(e)| > 1, i = 1, \dots, q\}$$

coordinatewise radial integration gives, for every non-negative measurable Γ on E

$$p_e \mathbb{E}\{\Gamma(\Theta)\} = \int_{A_e} \Gamma\left(\frac{f}{|f(e)|}\right) \nu_Z(df). \quad (61)$$

Taking $\Gamma = 1$ shows that $p_e = \nu_Z(A_e)$, and (61) shows that $\mathcal{L}(\Theta)$ is determined by ν_Z . The substitution $s = r \odot |Z(e)|$ in the radial integral yields $\nu_Z = p_e \nu_{\Theta}$, while $|\Theta(e)| = \mathbf{1}$ almost surely. For $t \in \mathbb{T}$ set

$$E_t = \{f \in E : \mathfrak{W}_f(t) > 0\}.$$

For a non-negative measurable H on E , define on E_e

$$F_H(f) = \mathfrak{W}_f(e) H\left(\frac{f}{|f(e)|}\right)$$

and extend it by zero elsewhere. Then $F_H \in \mathfrak{H}(E)$ and

$$F_H(B^h \Theta) = \mathbb{I}_{\{\mathfrak{W}_{\Theta}(h^{-1}) > 0\}} \mathfrak{W}_{\Theta}(h^{-1}) H\left(\frac{B^h \Theta}{|\Theta(h^{-1})|}\right), \quad F_H(\Theta) = H(\Theta)$$

hence (25) implies (29).

Conversely, assume (29). For $t \in \mathbb{T}$, take $h = t$ and $H = \mathbb{I}_{E_t^c}$. On $E_{t^{-1}}$, the path $B^t \Theta / |\Theta(t^{-1})|$ belongs to E_t . Hence (29) gives $\mathbb{P}\{\Theta \notin E_t\} = 0$. It follows that for every fixed $h \in \mathbb{T}$, $\mathfrak{W}_{\Theta}(h^{-1}) > 0$ almost surely. Every $F \in \mathfrak{H}(E)$ therefore satisfies

$$F(B^h \Theta) = \mathfrak{W}_{\Theta}(h^{-1}) F\left(\frac{B^h \Theta}{|\Theta(h^{-1})|}\right)$$

almost surely. Taking $H = F$ in (29) gives (25). Since multiplication of an intensity by p_e does not affect shift invariance, Theorem 3.1 completes the proof. $\square$

Proof of Proposition 3.3. Both exponentiated fields have unit anchor, so their represented intensities are σ -finite. For fixed h set

$$R_h f(t) = f(ht) - f(h), \quad U_h = e^{\langle \mathbf{1}, X(h) \rangle}, \quad V_h = e^{\langle \mathbf{1}, AX(h) \rangle}.$$

On paths anchored at zero, R_h is a measurable bijection with inverse $R_{h^{-1}}$ and commutes with A . Hence $\sigma(AR_h X) = \mathcal{G}_A$. Applying (29) with shift h^{-1} to e^X gives

$$\mathbb{E}\{U_h K(AR_h X)\} = \mathbb{E}\{K(AX)\}$$

for every non-negative cylinder-measurable K on $\mathcal{X}$; logarithms in the test functional may be extended measurably off positive values. In particular, $\mathbb{E}\{U_h\} = 1$. The same criterion for $\exp(AX)$ requires this identity with V_h in place of U_h . Since V_h is $\mathcal{G}_A$ -measurable, equality of the weighted tests is exactly (30); necessity also gives $\mathbb{E}\{V_h\} = 1$ by $K = 1$. If A is invertible, $\mathcal{G}_A = \sigma\{X(t) : t \in \mathbb{T}\}$, so the conditional identity reduces to (31). $\square$

Proof of Proposition 3.5. The unit anchor gives sigma-finiteness. If ν_Z is shift-invariant, apply (29) with $h = -1$ and $H(f) = g(-\log f(-1))$, extended measurably off positive values. For every non-negative Borel g on $\mathbb{R}^q$, this gives

$$\mathbb{E}\{g(L^-(1))\} = \mathbb{E}\left\{e^{\langle \mathbf{1}, L^+(1) \rangle} g(L^+(1))\right\},$$

since a Lévy process agrees almost surely with its left limit at a fixed positive time. This is (33).

Conversely, the unit-time measure identity includes $\mathbb{E}\{e^{\langle \mathbf{1}, L^+(1) \rangle}\} = 1$. The Esscher-transformed Lévy semigroup is therefore a probability semigroup; uniqueness of Lévy semigroups from their unit-time laws gives

$$\mathbb{P}\{L^-(t) \in dx\} = e^{\langle \mathbf{1}, x \rangle} \mathbb{P}\{L^+(t) \in dx\}, \quad t \geq 0.$$

Let P_t and P_t^* be the kernels of $x + L^+(t)$ and $x - L^-(t)$, then

$$e^{-\langle \mathbf{1}, x \rangle} dx P_t(x, dy) = e^{-\langle \mathbf{1}, y \rangle} dy P_t^*(y, dx).$$

This duality and independent increments give a stationary two-sided Poisson system with starting intensity $e^{-\langle \mathbf{1}, x \rangle} dx$. Exponentiation identifies its trajectory intensity with ν_Z . $\square$

Proof of Theorem 3.7. Write $\nu = \nu_Z$ and choose a finite, strictly positive measurable density g on G with $\int_G g d\vartheta = 1$. On the auxiliary space with measure $d\mathbb{P}(\omega)g(r)\vartheta(dr)$, let R be the radial coordinate and put $\mu = \mathcal{L}(S_R Z)$. For a real, finite-valued $F \in L^1(\nu)$ set

$$T_a F(f) = F(B^{a^{-1}} f), \quad U_F(a, \omega, r) = g(r)^{-1} T_a F(S_r Z(\omega)).$$

The field U_F is jointly measurable: each coordinate of $(a, \omega, r) \mapsto B^{a^{-1}} S_r Z(\omega)$ is $r_i Z_i(at, \omega)$, so this map is measurable into $(\mathcal{X}, \mathcal{E}_{\text{cyl}})$. Utilising (23) we have

$$\mathbb{E}_{\text{aux}}|U_F(a)| = \|F\|_{L^1(\nu)}, \quad \mathbb{E}_{\text{aux}}|U_F(a) - U_F(c)| = \|T_a F - T_c F\|_{L^1(\nu)}.$$

Since $T_{va} F = (T_a F) \circ B^{v^{-1}}$, (57)–(58) hold for U_F . Lemma 7.2 implies further

$$\|T_a F - F\|_{L^1(\nu)} \longrightarrow 0 \quad \text{as } a \rightarrow e. \quad (62)$$

The radial representation and strict positivity of g give $\mu \sim \nu$. In particular, $L^1(\nu)$ convergence implies convergence in μ -probability, by Markov's inequality and absolute continuity of the finite measure μ with respect to ν .

σ -finiteness provides a finite, everywhere strictly positive measurable $w \in L^1(\nu)$. Fix $u \in \mathbb{T}$ and $i \in \{1, \dots, q\}$, and put

$$\psi(x) = \frac{x}{1 + |x|}, \quad \psi_{i,u}(f) = \psi(f_i(u)).$$

Apply (62) to w and $w\psi_{i,u}$. Taking the ratio, whose limiting denominator is strictly positive yields

$$\psi(f_i(au)) = \frac{T_a(w\psi_{i,u})(f)}{T_a w(f)} \longrightarrow \frac{w(f)\psi_{i,u}(f)}{w(f)} = \psi(f_i(u)) \quad \text{in } \mu\text{-probability.}$$

The continuous inverse of $\psi : \mathbb{R} \rightarrow (-1, 1)$ therefore yields $R_i Z_i(au) \rightarrow R_i Z_i(u)$ in auxiliary probability. Dividing by the same finite positive random variable R_i gives $Z_i(au) \rightarrow Z_i(u)$ in probability. Since there are finitely many coordinates, $Z(au) \rightarrow Z(u)$ in probability. Finally, for $s \rightarrow t$, take $u = t$ and $a = st^{-1} \rightarrow e$. $\square$

Proof of Theorem 4.1. We first treat normalised log-Gaussian marks on $C(\mathbb{T}, \mathbb{R}^q)$. Write $P_k = \mathcal{L}(Z_k)$. For measures μ concentrated on strictly positive paths $f \in \mathcal{C}$ with $f(e) = \mathbf{1}$, write

$$A_h \mu(G) = \int_{\mathcal{C}} \mathfrak{W}_f(h^{-1}) \mathbb{I}_{\{B^h f / f(h^{-1}) \in G\}} \mu(df), \quad h \in \mathbb{T}, \quad G \in \mathcal{B}(\mathcal{C}).$$

Gaussian exponential moments give $c_k(h) := \mathbb{E}\{\mathfrak{W}_{Z_k}(h^{-1})\} \in (0, \infty)$, and the laws $P_k^{[h]} := c_k(h)^{-1} A_h P_k$ are again log-Gaussian with value $\mathbf{1}$ at e . Indeed, writing $X_k = \log Z_k$, the weight is $\exp\{\langle \mathbf{1}, X_k(h^{-1}) \rangle\}$. Gaussian tilting leaves the covariance unchanged and shifts the mean at t by $\text{Cov}\{X_k(t), X_k(h^{-1})\} \mathbf{1}$, while the log-field after normalisation is $X_k(h^{-1}t) - X_k(h^{-1})$. Direct substitution gives $A_h A_g = A_{hg}$ and $A_e = \text{id}$. Consequently, for log-Gaussian probability laws with the same anchor normalisation, the normalised transforms $P^{[h]} = A_h P / (A_h P)(\mathcal{C})$ satisfy $(P^{[g]})^{[h]} = P^{[hg]}$ and $(P^{[h]})^{[h^{-1}]} = P$. In particular, the laws $P_k^{[h]}$, $k = 1, \dots, n$, are distinct for every fixed h .

We first verify finite log-Gaussian identifiability on $\mathcal{C}$. Given a finite family of distinct log-Gaussian laws $Q_1, \dots, Q_\ell$, choose finitely many log evaluations whose joint Gaussian laws distinguish every pair. This is possible because the evaluations generate $\mathcal{B}(\mathcal{C})$ and the laws are concentrated on strictly positive paths. Write a_j and V_j for the means and covariance matrices of these joint laws. A vector v can be chosen so that all pairs $(\mu_j, \sigma_j^2) = (v^\top a_j, v^\top V_j v)$ are distinct: the exceptional vectors lie in a finite union of proper polynomial zero sets. If $\sum_{j=1}^\ell d_j Q_j = 0$, the moment generating functions of these scalar projections give

$$\sum_{j=1}^\ell d_j e^{r\mu_j + r^2 \sigma_j^2 / 2} = 0, \quad r \in \mathbb{R}.$$

Choose the largest variance and, among terms with that variance, the largest mean. Dividing by its exponential and letting $r \rightarrow \infty$ shows that its coefficient is zero. Repeating proves linear independence. The argument also permits singular or zero covariance matrices for the Gaussian log-fields.

Condition (29) makes stationarity equivalent to

$$\sum_{k=1}^n p_k c_k(h) P_k^{[h]} = \sum_{k=1}^n p_k P_k, \quad h \in \mathbb{T}. \quad (63)$$

Linear independence, applied to the union of the two finite families, and positivity of the coefficients give a unique permutation ϖ_h with $P_k^{[h]} = P_{\varpi_h(k)}$ and $p_k c_k(h) = p_{\varpi_h(k)}$. These are precisely the weighted identities in the statement. Their converse follows by summing over k and applying (29).

Sample continuity and Gaussianity imply continuity of the means and covariance kernels of the log-fields: Gaussian increments tending to zero in probability have means and covariances tending to zero. The tilting and normalisation formulas above therefore show that the log-means and log-covariances under $P_k^{[h]}$ depend continuously on h . Choose a finite log-evaluation map and a scalar projection distinguishing $P_1, \dots, P_n$ as above. Under (63), its mean-variance pair under $P_k^{[h]}$ is continuous in h and ranges over a finite set of distinct pairs. Hence $h \mapsto \varpi_h$ is continuous. The identity $(P_k^{[g]})^{[h]} = P_k^{[hg]}$ gives $\varpi_{hg} = \varpi_h \circ \varpi_g$ by distinctness, and $\varpi_e = \text{id}$. Thus ϖ is a continuous homomorphism. If $\mathbb{T}$ is connected, continuity gives $\varpi_h = \text{id}$ for every h . The weighted identities and (29) then give stationarity of every component system; the converse follows by linearity.

For unnormalised log-Gaussian components put $\gamma_k = \mathbb{E}\{\mathfrak{W}_{Z_k}(e)\} \in (0, \infty)$ and let Z_k^0 have the canonical law (28). Then Z_k^0 is log-Gaussian with $Z_k^0(e) = \mathbf{1}$ almost surely, and the single-anchor criterion and Corollary 3.2 give

$$\nu_{Z_k} = \gamma_k \nu_{Z_k^0}, \quad k = 1, \dots, n.$$

After dividing the weights $p_k \gamma_k$ by $\sum_{j=1}^n p_j \gamma_j$ and merging equal laws, the normalised result applies. Since $\nu_Z = \sum_{k=1}^n p_k \gamma_k \nu_{Z_k^0}$, this proves (35).

For (21), fix $m \geq 1$. Each log Z_k has continuous mean, bounded on K_m , and its centred restriction is a Gaussian random element of the separable Banach space $C(K_m, \mathbb{R}^q)$. Fernique's theorem gives

$$\mathbb{E} \left\{ \prod_{i=1}^q \rho_{m,i}(Z_k) \right\} = \mathbb{E} \left\{ \exp \left\{ \sum_{i=1}^q \sup_{t \in K_m} \log Z_{k,i}(t) \right\} \right\} < \infty, \quad k = 1, \dots, n.$$

A finite sum with weights p_k proves (21), and hence (19), completing the Gaussian case.

Now let $\mathbb{T} = \mathbb{R}$ and first suppose all marks are log-Lévy. We may work on the regular carrier $\mathcal{D}_+ = \{\exp x : x \in D(\mathbb{R}, \mathbb{R}^q)\}$, since all component intensities are concentrated there. Write $L_k^\pm$ for the processes in (32) for Z_k , and $\varphi_k^\pm$ for the characteristic functions of $L_k^\pm(1)$. The unit-time laws determine the Lévy laws and, by independence of the halves, the law of Z_k . Thus the pairs $(\varphi_k^+, \varphi_k^-)$ are distinct. For $u \in \mathbb{Q}^q$ and $f \in \mathcal{D}_+$ put

$$g_u(f, s) = \exp\{i\langle u, \log f(s+1) - \log f(s) \rangle\}.$$

For $N \in \mathbb{N}$ define

$$A_N^+(f, u) = \frac{1}{N} \int_0^N g_u(f, s) ds, \quad A_N^-(f, u) = \frac{1}{N} \int_{-N}^0 g_u(f, s) ds.$$

Joint evaluation makes these maps measurable. For fixed $r \in [0, 1)$, the unit increments at $n+r$ are independent and identically distributed on either half, apart from the single

interval crossing zero. The strong law, Fubini's theorem and bounded convergence in r imply simultaneously for $u \in \mathbb{Q}^q$

$$A_N^\pm(Z_k, u) \longrightarrow \varphi_k^\pm(u) \quad \text{almost surely.}$$

Let E_k be the measurable set on which these limits hold with the pair $(\varphi_k^+, \varphi_k^-)$. These sets are disjoint and scale invariant. They are also invariant under every shift: translating an integration interval by h changes its average by at most $2|h|/N$. By (11), ν_{Z_k} is concentrated on E_k , so

$$\nu_Z|_{E_k} = p_k \nu_{Z_k}.$$

If ν_Z is shift-invariant, its restriction to the invariant set E_k is shift-invariant. This proves the necessity in (35); sufficiency follows by linearity.

Finally, allow both log-Gaussian and log-Lévy marks on the càdlàg carrier. The set of continuous paths is measurable and invariant under shifts and scalings. A log-Lévy mark whose two halves have zero Lévy measures is log-Gaussian; every other log-Lévy mark is discontinuous somewhere on $\mathbb{R}$ almost surely. Restricting a stationary mixture to discontinuous paths and then to the sets E_k isolates each such log-Lévy component. The continuous restriction, if nonzero, is a finite log-Gaussian mixture; the connected Gaussian case proves (35). $\square$

Proof of Lemma 5.2. Radial integration gives (44) and χ -homogeneity. Since $Q_i \not\equiv 0$ almost surely for $i = 1, \dots, q$, the measure ν_Q^{cl} satisfies (56); its total mass is $\int_{\mathbb{T}} \vartheta(G) \lambda(dv) = \infty$. Condition (19) and Lemma 7.1 therefore give $\nu_Q^{\text{cl}} \in M_\sigma(\mathcal{C})$. The marking and mapping theorems identify (43) as a Poisson random measure with this intensity. Finally, $B^a B^v = B^{av}$ and left invariance of λ make the intensity shift-invariant, so Theorem 3.1 proves stationarity. $\square$

Proof of Proposition 5.4. Condition (46) implies that no component of Z is an identically zero path. Thus (16) gives (17), and Lemma 7.1(ii) gives $\nu \in M_\sigma(\mathcal{C})$. Choose n with $e \in K_n$. Radial integration and (19) give

$$p = \nu\{f \in \mathcal{C} : |f_i(e)| > 1, i = 1, \dots, q\} \leq \nu(U_{(n, \dots, n), 1}) < \infty.$$

Shift invariance gives $\mathbb{E}\{\mathfrak{W}_Z(h)\} = p$ for every $h \in \mathbb{T}$. If $p = 0$, Tonelli's theorem would give $\mathbb{E}\{\mathcal{S}_\Pi(Z)\} = 0$, contrary to (46). Thus $p > 0$ and $\mathbb{P}^*$ is well defined.

Since $\mathbb{P}^* \ll \mathbb{P}$, condition (46) holds also under $\mathbb{P}^*$. Moreover, $\mathfrak{W}_Z(e) > 0$ $\mathbb{P}^*$ -almost surely, so Q^{com} is a well-defined $\mathcal{C}$ -valued random element satisfying (17). The scaling vector in (47) has character $p/\mathcal{S}_\Pi(Z)$, hence $\mathcal{S}_\Pi(Q^{\text{com}}) = p$ $\mathbb{P}^*$ -almost surely.

Write $Q = Q^{\text{com}}$. For $F \in \mathfrak{H}(\mathcal{C})$, put $G_F = F/\mathcal{S}_\Pi$ on $\{0 < \mathcal{S}_\Pi < \infty\}$ and $G_F = 0$ elsewhere. This is measurable and scale invariant. We use $0 \cdot \infty = 0$. For fixed v , the functional $f \mapsto \mathfrak{W}_f(e)G_F(B^v f)$ belongs to $\mathfrak{H}(\mathcal{C})$. Applying (25) with shift v^{-1} and then Tonelli's theorem gives

$$\begin{aligned} \int_{\mathbb{T}} \mathbb{E}^* F(B^v Q) \lambda(dv) &= \int_{\mathbb{T}} \mathbb{E}\{\mathfrak{W}_Z(e)G_F(B^v Z)\} \lambda(dv) \\ &= \int_{\mathbb{T}} \mathbb{E}\{\mathfrak{W}_Z(v)G_F(Z)\} \lambda(dv) \\ &= \mathbb{E}\{\mathcal{S}_\Pi(Z)G_F(Z)\} = \mathbb{E}\{F(Z)\}. \end{aligned}$$

Taking $F = \prod_{i=1}^q \rho_{K_i, i}$ and using (21) and cofinality yields (41). Taking $F(f) = \int_G \mathbb{I}_{\{S_r f \in A\}} \vartheta(dr)$, $A \in \mathcal{E}_{\mathcal{C}}$, yields (48) by (42). $\square$

Proof of Theorem 6.1. By Proposition 5.4, $Q = Q^{\text{com}}$ under $\mathbb{P}^*$ is a product cluster, $\mathcal{S}_{\Pi}(Q) = p$ almost surely, and $\nu = \nu_Q^{\text{cl}}$. A shifted cluster hits K at level y precisely when its shift belongs to $KE_y(Q)^{-1}$. If $D_K \subset K$ is countable and dense, continuity gives

$$\{v \in \mathbb{T} : v \in KE_y(Q)^{-1}\} = \bigcup_{d \in D_K} \bigcap_{i=1}^q \{v \in \mathbb{T} : Q_i(v^{-1}d) > y_i\},$$

which also establishes the required joint measurability in (v, y, Q) . Hence

$$H^\nu(K) = \mathbb{E}^* \left\{ \int_{(0, \infty)^q} \lambda(KE_y(Q)^{-1}) dy \right\}. \quad (64)$$

Choose a compact symmetric identity neighbourhood V and put $U = V^2$. Condition (41) implies

$$\mathbb{E}^* \left\{ \int_{(0, \infty)^q} \lambda(UE_y(Q)^{-1}) dy \right\} \leq \Lambda_{(U, \dots, U)}(Q) < \infty. \quad (65)$$

Indeed, $v \in UE_y(Q)^{-1}$ implies $y_i < \rho_{U,i}(B^v Q)$ for $i = 1, \dots, q$; integrate this inclusion first over y and then over v . If $E_y(Q)$ were not relatively compact, it would contain points t_j for which the sets Ut_j^{-1} are pairwise disjoint. Unimodularity makes every such set have measure $\lambda(U) > 0$ contradicting (65). Thus $E_y(Q)$ is relatively compact for $dy \otimes \mathbb{P}^*$ -almost every (y, Q) . For a nonempty relatively compact open set E , fix $t \in E$ and a compact $C \supset E^{-1} \cup \{e\}$. Right invariance and (50) give

$$1 = \frac{\lambda(F_n t^{-1})}{\lambda(F_n)} \leq \frac{\lambda(F_n E^{-1})}{\lambda(F_n)} \leq \frac{\lambda(F_n C)}{\lambda(F_n)} \rightarrow 1.$$

The ratio is zero for $E = \emptyset$. For domination, choose a maximal disjoint family $(a_j V)_{j=1}^{N_n}$ with $a_j \in F_n$. Then $F_n \subset \bigcup_{j=1}^{N_n} a_j U$ and $N_n \lambda(V) \leq \lambda(F_n V)$, so

$$\frac{\lambda(F_n E^{-1})}{\lambda(F_n)} \leq \frac{\lambda(F_n V)}{\lambda(F_n) \lambda(V)} \lambda(U E^{-1}) \leq c_V \lambda(U E^{-1}),$$

where $c_V < \infty$ follows from (50). Dominated convergence in (64), using (65), yields $\mathcal{H}^\nu = \mathbb{E}^* \{\mathcal{R}_{\text{sup}}(Q)\} < \infty$. Since $\mathcal{S}_{\Pi}(Q) = p > 0$, there is common positivity somewhere, so $\mathcal{R}_{\text{sup}}(Q) > 0$ almost surely. This proves strict positivity as well as independence of the windows. Finally, $\mathcal{R}_{\text{sup}}(S_r f) = \chi(r) \mathcal{R}_{\text{sup}}(f)$. The definition of Q^{com} and the density of $\mathbb{P}^*$ therefore give the second equality in (53).

For the reciprocal representation, write $A = \{f : f(e) > \mathbf{1}\}$. Under the substitution $y_i = r_i^{-1}$, the excursion set of $S_r B^v Q$ is $v E_y(Q)$, whose Haar measure is $\lambda(E_y(Q))$. Moreover, $S_r B^v Q \in A$ exactly when $v^{-1} \in E_y(Q)$. Inversion preserves Haar measure, and each nonempty $E_y(Q)$ has positive finite measure: positivity follows from continuity, while

$$\lambda(E_y(Q)) \leq \frac{\mathcal{S}_{\Pi}(Q)}{\prod_{i=1}^q y_i} < \infty.$$

Consequently, radial substitution and Tonelli's theorem give

$$\int_A \frac{\nu(df)}{\tau(f)} = \mathbb{E}^* \left\{ \int_{(0, \infty)^q} \frac{\lambda(E_y(Q)^{-1})}{\lambda(E_y(Q))} dy \right\} = \mathbb{E}^* \{\mathcal{R}_{\text{sup}}(Q)\},$$

where the fraction is zero for the empty excursion. By the definition of Θ , $\Theta(e) = \mathbf{1}$ almost surely. For every nonnegative measurable G , radial substitution gives

$$\int_A G(f) \nu(df) = p\mathbb{E}\left\{\int_{(1,\infty)^q} G(S_r\Theta) \vartheta(dr)\right\} = p\mathbb{E}\{G(S_R\Theta)\}.$$

Taking $G = 1/\tau$ proves the last equality in (54). $\square$

Proof of Corollary 6.2. Write $S = S_{\kappa,V}$ and $c = \mathbf{1}^\top V \mathbf{1} > 0$. The singleton Gaussian criterion has $\Sigma(t, s) = S(t) + S(-s) - S(t - s)$ and $b(t) = \frac{1}{2}\{S(-t) - S(t)\}\mathbf{1}$. Since $S(-t) = S(t)^\top$, direct substitution verifies $\langle b(t), \mathbf{1} \rangle = 0$ and (37)–(38). Thus ν_Z is shift-invariant, and the Gaussian part of Theorem 4.1 gives (19). Moreover we have

$$\mathfrak{W}_Z(t) = \exp\{\mathbf{1}^\top Y_{\kappa,V}(t) - c|t|^\kappa\}.$$

The Gaussian process $\mathbf{1}^\top Y_{\kappa,V}$ has the law of $\sqrt{2c} B_{\kappa/2}$, where $B_{\kappa/2}$ is two-sided fractional Brownian motion, with the linear Gaussian interpretation for $\kappa = 2$. Gaussian tail bounds at integer times and on unit increments, followed by Borel–Cantelli, give $B_{\kappa/2}(t) = o(|t|^\kappa)$ almost surely as $|t| \rightarrow \infty$. Hence $0 < \mathcal{S}_\Pi(Z) < \infty$ almost surely. For every positive sequence $T_n \rightarrow \infty$, the windows $F_n = [0, T_n]$ satisfy (50). Applying Theorem 6.1 and substituting $y_i = e^{x_i}$ in (52) yields (7), with $\mathcal{H} = \mathcal{H}^{\nu_Z}$. Since $Z(0) = \mathbf{1}$, (53) and (54) give (55). Finally, continuity and $R_i > 1$ give $\tau(S_R Z) > 0$, while $\tau(S_R Z) \leq \chi(R)\mathcal{S}_\Pi(Z) < \infty$ almost surely. $\square$

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